Project Writeup

MULTIMODAL CONTRADICTION AND BEHAVIOURAL ANALYSIS IN HUMAN CONVERSATIONS USING DIALOGUE GRAPHS AND FACIAL ACTION UNITS

Rithika M (23MIC0072)

M.Tech Integrated in Computer Science and Engineering, Vellore Institute of Technology (VIT), Vellore

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1. Title

Multimodal Contradiction and Behavioural Analysis in Human Conversations Using Dialogue Graphs and Facial Action Units.

The project studies conversational contradiction (a relation between claims made at different points in a dialogue) using transcript processing, typed dialogue graphs, and time-aligned facial and body descriptors. Its NLP focus is utterance meaning in conversational context; visual behaviour supplies contextual information and is *not* treated as evidence of dishonesty.

2. Motivation

Human conversations unfold through short, incomplete and overlapping turns. A later statement may repeat an earlier claim, add detail, change the subject or explicitly withdraw what was said before. Contradictions can be separated by intervening turns and can depend on *who* made an earlier claim. This motivates a representation that preserves dialogue order and speaker identity instead of flattening the conversation into one undifferentiated string.

A graph exposes those relations directly, while a plain text baseline provides a necessary check on whether the added structure is actually useful. The project also examines a practical obstacle in multimodal NLP: extraction completeness and temporal alignment decide whether later experiments are meaningful at all. A model trained on a truncated video file, or on features of the listener rather than the speaker, would answer a different question from the intended one. Coverage audits, missing-value masks and explicit exclusion statuses are therefore part of the methodology.

3. Existing Gap / Event That Motivated the Project

The following gaps shaped the project:

- **Text-only contradiction work.** Dialogue NLI [1] and DECODE [2] model pairwise consistency in dialogue but have no facial or body branch and do not combine textual relations with audited visual summaries.
- **Veracity is not contradiction.** The Box of Lies corpus [3] labels game phases as lie or truthful. That is a different target and cannot be converted into semantic contradiction labels, so the project keeps the

two label spaces strictly separate.

- **Tools are not honesty classifiers.** OpenFace [4] and MediaPipe [5] estimate visible behaviour only; they do not measure truth, intention or mental state.
- **Conversation structure.** Relational graph convolution [6] and DialogueGCN [7] motivate speaker/context relations, but emotion recognition is a different target from contradiction.
- **Data and extraction reality.** The project folder holds 49 MP4 files but only 25 EAF annotation files, and the video-to-annotation mapping is unverified. Of 49 OpenFace outputs, 48 fail the duration-based completeness audit (median coverage 3.42%). Early multi-face batch runs stopped after the first seconds of many videos.
- **Annotation scarcity.** Only one contradiction was verified in 59 reviewed pairs, so contradiction-class training and evaluation are not yet possible.

The project opportunity is therefore to link textual claims, dialogue relations and audited visual summaries while retaining separate label spaces, and to document honestly what the current data can and cannot support.

4. Architecture

EAF and video inputs feed parsing, quality checks and temporal alignment. Time is the join key, and semantic labels are kept outside the visual extraction path. A failure in extraction or an ambiguous label boundary produces an explicit status rather than an assumed value, so downstream omissions are inspectable.

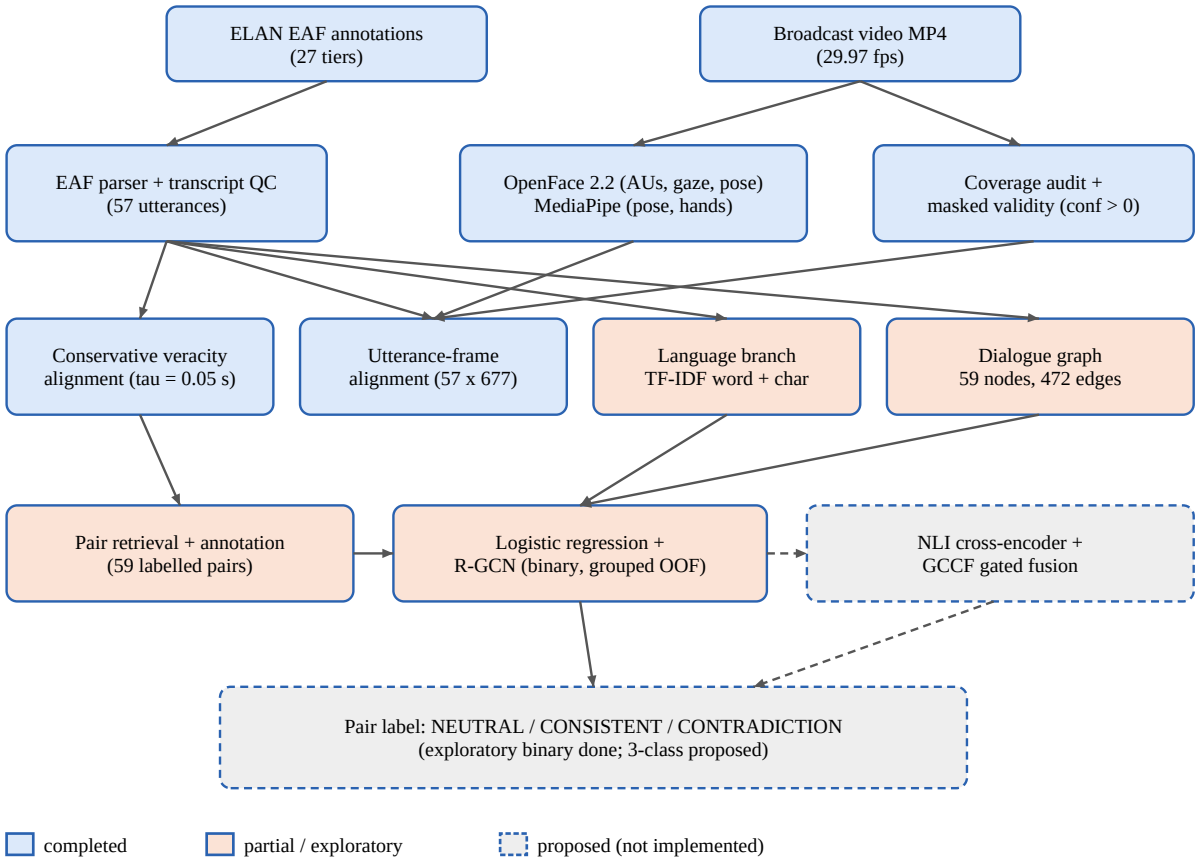

Figure 1. Pipeline architecture with completed, partial and proposed components.

Key architectural stages:

- **Input layer:** ELAN EAF annotation file and broadcast video (29.97 fps, 640 x 360).

- **Pre-processing and alignment:** EAF parsing with TIME_SLOT resolution, transcript QC, conservative veracity alignment and utterance-frame alignment.
- **Feature branches:** language (TF-IDF), dialogue graph (R-GCN encoder), facial branch (OpenFace AUs, gaze, head pose) and body branch (MediaPipe pose and hands).
- **Fusion and output:** exploratory early concatenation with logistic regression (binary only). Graph-conditioned contradiction fusion (GCCF) with a gated sigmoid combination of text and graph experts and a three-class softmax is specified but **proposed, not trained**.

5. Complete Tech Stack

Layer	Tool / version	Purpose
Language and platform	Python 3.13; Google Colab notebook (Video_detection.ipynb, 549 code cells); Google Drive storage; Ubuntu 24.04.1 LTS	Development and persistence of intermediate outputs
Hardware	Tesla T4 available; R-GCN experiments ran on CPU	Compute (free tier)
Build toolchain	g++ 13.3.0; CMake 3.31.10; system OpenCV 4.6.0; patched dlib 19.13; Boost filesystem	Compiling OpenFace from source
Facial analysis	OpenFace 2.2.0 (FeatureExtraction with -aus -gaze -pose)	17 AU intensities, 18 AU presences, 8 gaze, 6 head-pose, 280 eye-landmark columns per frame
Body and hands	MediaPipe 1.1.1 (pose_landmarker_lite, hand landmarker, VIDEO mode, every 3rd frame)	33 pose landmarks (x, y, z, visibility); 21 landmarks per hand, up to four hands
Annotation format	ELAN EAF (XML) parsed with ElementTree	Speech tiers and veracity tier
Data handling	pandas, NumPy	Tables, alignment, aggregation
Classical ML	scikit-learn (TfidfVectorizer, class-balanced LogisticRegression, StratifiedGroupKFold)	Text baseline and grouped out-of-fold evaluation
Graph deep learning	PyTorch 2.11.0; custom RelationalGraphConv (R-GCN) layer	Relation-specific message passing over the dialogue graph
NLP utilities	spaCy	Shared named-entity search
Version control	GitHub	Source code

Model settings. Random seed 42. TF-IDF: lower-cased word unigrams and bigrams, sublinear term frequency, smoothed IDF, L2 normalisation. R-GCN: two layers (516 to 128 to 64), ReLU, dropout 0.2; pair vector $[h_i; h_j; |h_i - h_j|; h_i * h_j]$ (256-d) into a 256 to 64 to 2 perceptron with dropout 0.25; 100 epochs per fold, Adam, learning rate 0.003, weight decay 1e-4, inverse-frequency class weights.

6. Datasets Used

Box of Lies [3] consists of two-person game conversations in which one participant describes a hidden object and the other guesses whether the description is truthful. The original corpus has 25 videos and 26 unique participants. The project folder contains 49 MP4 files and an annotation archive with 25 EAF files (no video); the mapping of the extra MP4 files to annotations is unverified. The only conversation processed in full is **video01.mp4 with 1.BoL.An.Tr.eaf**.

Artefact	Size	Interpretation
video01 source	375.008 s; 11,239 frames; 640 x 360; about 29.97 fps	Only conversation processed in full

Artefact	Size	Interpretation
EAF annotation	1,614 time slots; 27 tiers; 807 time-aligned annotations	Speech from Host_verbal and Guest_verbal
Parsed speech	105 raw rows (48 empty) to 57 utterances	30 Guest, 27 Host; 54.251 to 344.300 s; 48 utterances overlap another
Veracity tier	11 intervals	Seven lie, four truthful
OpenFace output	11,239 x 334	Frame-level facial and eye descriptors; 99.99% time coverage
Sampled multimodal merge	3,747 x 496	All samples matched within 0.05 s (max 0.001 s)
Validated utterance table	57 x 677	658 aggregate features plus metadata and QC
Reviewed semantic pairs	59	51 NEUTRAL; 7 CONSISTENT; 1 CONTRADICTION (single annotator)

Semantic pair annotations were created by the authors: candidates came from evenly spaced pairs, focused claim searches and TF-IDF retrieval, and each retained label records a reason. A positive-label audit downgraded five of eleven provisional positives. The only verified contradiction links an earlier description naming a Backstreet Boys referent to a later explicit correction naming John Stamos.

7. Prompts Used for Construction of the Website

Not applicable. The developed product is a Colab notebook workflow producing CSV outputs, graphs and figures, not a deployed website or application interface. The report does not record any prompts used to construct a website, and none are listed here to avoid inventing content. If prompts were used for any assistance during development, they should be added by the authors from their own records.

8. NLP Pipeline

The pipeline runs in the following ordered stages (done / partial / proposed status in brackets):

- **Step 1 - Parsing [done].** Resolve ELAN TIME_SLOT identifiers to seconds, read time-aligned annotations, select speech and veracity tiers separately, strip empty strings and exact duplicates, sort by start and end time, and assign stable utterance IDs. Overlap and genuinely repeated turns are kept as conversational properties.
- **Step 2 - Transcript QC [done].** Counts of missing fields, reversed or invalid timestamps, duration mismatches and times beyond the video are all zero in the cleaned transcript.
- **Step 3 - Visual extraction and audit [done for video01].** OpenFace per-frame output and MediaPipe sampled pose/hand records; duration-based completeness audit. Frames are valid when success = 1 and confidence > 0 (8,405 of 11,239 valid).
- **Step 4 - Conservative veracity alignment [done].** Overlap of each utterance with lie/truthful intervals is summed per label. One label gives a direct assignment; both labels with minor overlap <= 0.05 s keep the dominant label; larger cross-label overlap is excluded. Result: 48 lie, 4 truthful, 5 excluded. These labels describe game intervals, not semantic contradiction.
- **Step 5 - Utterance feature tables [done].** Masked mean and standard deviation of 329 facial columns (658 features) with detection rate and quality group (HIGH / MODERATE / LOW / NO_FACE).
- **Step 6 - Candidate retrieval [partial].** Each target is compared only with earlier utterances. Ranking score = 0.5 x word cosine + 0.3 x character 3-5-gram cosine + 0.2 x Jaccard of content words. This prioritises reading candidates and is not a contradiction probability.

- **Step 7 - Pair annotation [partial].** Human labelling as NEUTRAL, CONSISTENT or CONTRADICTION with recorded rationale; 1,596 chronological pairs reduced to 59 reviewed pairs.
- **Step 8 - Text representation [done].** Pair text is formatted as PREVIOUS STATEMENT / TARGET STATEMENT and encoded with TF-IDF; class-balanced logistic regression predicts CONSISTENT vs NEUTRAL, with the vectoriser fitted inside the training fold.
- **Step 9 - Dialogue graph [done].** 57 statement nodes plus 2 speaker nodes (59 total). Six active relation types and 472 directed edges: similar_topic 136, same_speaker 110, follows 56, followed_by 56, spoken_by 57, has_statement 57. The shared_entity relation is defined but has zero instances.
- **Step 10 - Relational encoder [done, exploratory].** Two-layer R-GCN over the graph, then a pair classifier. Evaluation is transductive (test text is visible in the graph; only the loss is restricted to training labels).
- **Step 11 - Evaluation [done, exploratory].** StratifiedGroupKFold (3 folds, grouped by target utterance, seed 42), pooled out-of-fold metrics with a majority-class baseline.
- **Step 12 - Proposed extensions.** Pre-trained NLI cross-encoder, GCCF gated fusion (g = sigmoid(W[h_text; h_struct] + b); fused = g * h_text + (1 - g) * h_struct), masked visual expert after speaker-to-face verification, and three-class prediction.

Results summary (58 binary pairs: 7 CONSISTENT, 51 NEUTRAL)

Model	Accuracy	Macro-F1	Weighted F1	CONSISTENT F1
Always NEUTRAL	0.8793	0.4679	0.8228	0.000
TF-IDF + logistic regression	0.8448	0.6554	0.8492	0.400
R-GCN (transductive)	0.8793	0.6485	0.8646	0.364

Both learned models exceed the majority baseline on macro-F1 but neither exceeds its accuracy. With seven minority examples one prediction materially changes the score, so the outcome supports pipeline feasibility and a documented baseline comparison, not graph superiority or generalisation to unseen conversations.

Known limitations

- One conversation, one annotator, one contradiction; no inter-annotator agreement.
- Speaker-to-face attribution is unresolved: all 57 utterances remain NOT_VERIFIED, so visual features may describe a listener. Visual-only and early-fusion runs reproduced the TF-IDF confusion matrix and provide no evidence that behaviour helps.
- Lexical retrieval favours repetition and can miss paraphrased contradictions. Leave-one-conversation-out evaluation was not run.

9. Users, Inputs and Outputs

Users

The system is a research workflow, not a public application. Intended users are:

- **NLP / computational-linguistics researchers and students** studying dialogue contradiction and multimodal conversation analysis.
- **Annotators and reviewers** who use the retrieved candidate pairs and context sheets to assign semantic labels.
- **Future project contributors** extending extraction to more conversations or training a transformer / GCCF model.

The design does not support claims about any individual's honesty or internal state, and any new data collection would need consent and ethical review.

Inputs (types and formats)

Input	Format	Details
Conversation video	MP4	About 29.97 fps, 640 x 360; video01 is 375.008 s (11,239 frames)
Time-aligned annotations	ELAN EAF (XML)	TIME_SLOT and tier structure; speech tiers Host_verbal and Guest_verbal; Veracity_description tier
Pair annotation labels	CSV	Earlier utterance ID, target utterance ID, label (NEUTRAL / CONSISTENT / CONTRADICTION), rationale, review state
Configuration	Python parameters	Seed 42; alignment tolerance 0.05 s; top_k = 5; similarity thresholds; fold settings

Outputs

Output	Format	Description
Cleaned utterance table	CSV	57 utterances with ID, speaker, start, end, duration, text
Frame-level facial features	CSV	OpenFace 11,239 x 334; sampled MediaPipe merge 3,747 x 496
Validated utterance feature table	CSV	57 x 677 (658 masked aggregates plus metadata and QC status)
Veracity alignment table	CSV	Clean label (lie / truthful / None) with status such as source_single_label, resolved_tiny_overlap or excluded_crosses_boundary
Pair table	CSV	59 labelled pairs with rationale and review state
Candidate lists	CSV	Ranked retrieval candidates (270 at top_k = 5) and review sheets
Dialogue graph	Node IDs and typed edge lists	59 nodes, 472 directed edges, six active relation types
Model predictions and metrics	Tables / confusion matrices	Pooled out-of-fold accuracy, macro-F1, weighted F1, per-class scores
Figures	Charts / diagrams	Coverage audit, modality availability, edge counts, confusion matrices
Pair prediction (model output)	Label	Currently CONSISTENT vs NEUTRAL; three-class output is future work

10. References

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